KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5081 | 5 Credits: 4 | Program Core | Source-grounded

Industrial Instrumentation

 To insight the measurement techniques of Viscosity, density, Specific gravity, and pH

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5081 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5081.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Instrumentation Engineering
Course code	5081
Course title	Industrial Instrumentation
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L: 3 T: 1 P: 0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

8 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To insight the measurement techniques of Viscosity, density, Specific gravity, and pH
- measurements.
- To explain different methods for measurement of physical parameters like humidity,
- moisture and radiation measurements.
- To explain different methods for measurement of physical parameters like force,
- To insight the measurement techniques of acceleration, vibration, thickness, and
- miscellaneous measurements.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding density, Specific gravity, and pH measuring instruments. Explain the construction, working principle and	See official syllabus
CO2	applications of humidity, moisture and radiation 14 Understanding measuring instruments. Apply different types of transducers for	See official syllabus
CO3	measurement of physical parameters like force, 14 Applying torque, speed and velocity measurements Compare and apply different types of transducers	See official syllabus
CO4	for acceleration, vibration, thickness, and 15 Applying miscellaneous measurements	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	different types of Viscosity, density, Specific gravity and pH measuring	3	As specified
Module 2	humidity, moisture and radiation measuring instruments.	1	As specified
Module 3	parameters like force, torque, speed and velocity measurements	1	As specified
Module 4	vibration, thickness, and miscellaneous measurements.	3	As specified

MODULE 1

different types of Viscosity, density, Specific gravity and pH measuring

This module is presented from the official syllabus scope for Industrial Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: different types of Viscosity, density, Specific gravity and pH measuring
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

density, Specific gravity, and pH 5 Understanding measurement Explain the construction and working principle of different types of Viscosity,

density, Specific gravity, and pH 5 Understanding measurement Explain the construction and working principle of different types of Viscosity, is an official topic in Module 1 of Industrial Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify density, Specific gravity, and pH 5 Understanding measurement Explain the construction and working principle of different types of Viscosity, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, density, specific gravity, and ph 5 understanding measurement explain the construction and working principle of different types of viscosity, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What density, Specific gravity, and pH 5 Understanding measurement Explain the construction and working principle of different types of Viscosity, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ density, Specific gravity, and pH 5 Understanding measurement Explain the construction and working principle of different types of Viscosity, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

5 Understanding density, Specific gravity, and pH measuring instruments. Compare and apply different transducers for

5 Understanding density, Specific gravity, and pH measuring instruments. Compare and apply different transducers for is an official topic in Module 1 of Industrial Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding density, Specific gravity, and pH measuring instruments. Compare and apply different transducers for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding density, specific gravity, and ph measuring instruments. compare and apply different transducers for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding density, Specific gravity, and pH measuring instruments. Compare and apply different transducers for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 1-□□ 5 Understanding density, Specific gravity, and pH measuring instruments. Compare and apply different transducers for □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Viscosity, density, Specific gravity, and pH 5 Understanding measurements Viscosity measurement- Absolute viscosity - Kinematic viscosity - relative viscosity – units- capillary viscometer- Saybolt Viscometer - Redwood Viscometer - Applications of viscometer. Density and Specific gravity measurement- Definition- hydrometer- density measurement using LVDT- force balance method- differential pressure method. pH Measurements -Sorensen's scale–electrodes for pH measurements- Hydrogen electrode, glass electrode- calomel electrode, combined pH electrode- Digital pH meter. Explain the construction, working principle and applications of

Viscosity, density, Specific gravity, and pH 5 Understanding measurements Viscosity measurement- Absolute viscosity - Kinematic viscosity - relative viscosity – units- capillary viscometer- Saybolt Viscometer - Redwood Viscometer - Applications of viscometer. Density and Specific gravity measurement- Definition- hydrometer- density measurement using LVDT- force balance method- differential pressure method. pH Measurements -Sorensen's scale–electrodes for pH measurements- Hydrogen electrode, glass electrode- calomel electrode, combined pH electrode- Digital pH meter. Explain the construction, working principle and applications of is an official topic in Module 1 of Industrial Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Viscosity, density, Specific gravity, and pH 5 Understanding measurements Viscosity measurement- Absolute viscosity - Kinematic viscosity - relative viscosity – units- capillary viscometer- Saybolt Viscometer - Redwood Viscometer - Applications of viscometer. Density and Specific gravity measurement- Definition- hydrometer- density measurement using LVDT- force balance method- differential pressure method. pH Measurements -Sorensen's scale–electrodes for pH measurements- Hydrogen electrode, glass electrode- calomel electrode, combined pH electrode- Digital pH meter. Explain the construction, working principle and applications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, viscosity, density, specific gravity, and ph 5 understanding measurements viscosity measurement- absolute viscosity - kinematic viscosity - relative viscosity – units- capillary viscometer- saybolt viscometer - redwood viscometer - applications of viscometer. density and specific gravity measurement- definition- hydrometer- density measurement using lvdt- force balance method- differential pressure method. ph measurements -sorensen's scale–electrodes for ph measurements- hydrogen electrode, glass electrode- calomel electrode, combined ph electrode- digital ph meter. explain the construction, working principle and applications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Viscosity, density, Specific gravity, and pH 5 Understanding measurements Viscosity measurement- Absolute viscosity - Kinematic viscosity - relative viscosity – units- capillary viscometer- Saybolt Viscometer - Redwood Viscometer - Applications of viscometer. Density and Specific gravity measurement- Definition- hydrometer- density measurement using LVDT- force balance method- differential pressure method. pH Measurements -Sorensen's scale–electrodes for pH measurements- Hydrogen electrode, glass electrode- calomel electrode, combined pH electrode- Digital pH meter. Explain the construction, working principle and applications of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Viscosity, density, Specific gravity, and pH 5 Understanding measurements Viscosity measurement- Absolute viscosity - Kinematic viscosity - relative viscosity – units- capillary viscometer- Saybolt Viscometer - Redwood Viscometer - Applications of viscometer. Density and Specific gravity measurement- Definition- hydrometer- density measurement using LVDT- force balance method- differential pressure method. pH Measurements -Sorensen's scale–electrodes for pH measurements- Hydrogen electrode, glass electrode-calomel electrode, combined pH electrode- Digital pH meter. Explain the construction, working principle and applications of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
different types of Viscosity, density, Specific gravity and pH measuring
             instruments
             Understand the fundamentals of Viscosity,
  M1.01      density, Specific gravity, and pH                5
Understanding
             measurement
             Explain the construction and working
             principle of different types of Viscosity,
  M1.02                                                       5
Understanding
             density, Specific gravity, and pH measuring
             instruments.
             Compare and apply different transducers for
  M1.03      Viscosity, density, Specific gravity, and pH      5

Understanding
             measurements

Contents:
Viscosity measurement- Absolute viscosity - Kinematic viscosity - relative
viscosity –
units- capillary viscometer- Saybolt Viscometer - Redwood Viscometer -
Applications of
viscometer.
Density and Specific gravity measurement- Definition- hydrometer- density
measurement
using LVDT- force balance method- differential pressure method.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

humidity, moisture and radiation measuring instruments.

This module is presented from the official syllabus scope for Industrial Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: humidity, moisture and radiation measuring instruments.
- Official duration: As specified
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.

5 Understanding moisture and radiation measurement Explain the construction and working principle of different types of humidity, 5 Understanding M2.02 moisture and radiation measuring techniques Compare and apply different transducers for Understanding M1.03 humidity, moisture and radiation 4 measurement Humidity measurement-Definition - relative humidity –Absolute humidity - Dew Point- Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers. Moisture measurement – measurement of moisture in gases and liquids- capacitance hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges. Radiation measurement- Types of radiations- Geiger Muller tube- ionization chamber- scintillation counter- applications of radiation detectors. Apply different types of transducers for measurement of physical

5 Understanding moisture and radiation measurement Explain the construction and working principle of different types of humidity, 5 Understanding M2.02 moisture and radiation measuring techniques Compare and apply different transducers for Understanding M1.03 humidity, moisture and radiation 4 measurement Humidity measurement-Definition - relative humidity –Absolute humidity - Dew Point- Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers. Moisture measurement – measurement of moisture in gases and liquids- capacitance hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges. Radiation measurement- Types of radiations- Geiger Muller tube- ionization chamber- scintillation counter- applications of radiation detectors. Apply different types of transducers for measurement of physical is an official topic in Module 2 of Industrial Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding moisture and radiation measurement Explain the construction and working principle of different types of humidity, 5 Understanding M2.02 moisture and radiation measuring techniques Compare and apply different transducers for Understanding M1.03 humidity, moisture and radiation 4 measurement Humidity measurement-Definition - relative humidity – Absolute humidity - Dew Point- Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers. Moisture measurement – measurement of moisture in gases and liquids- capacitance hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges. Radiation measurement- Types of radiations- Geiger Muller tube- ionization chamber- scintillation counter- applications of radiation detectors. Apply different types of transducers for measurement of physical using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding moisture and radiation measurement explain the construction and working principle of different types of humidity, 5 understanding m2.02 moisture and radiation measuring techniques compare and apply different transducers for understanding m1.03 humidity, moisture and radiation 4 measurement humidity measurement-definition - relative humidity –absolute humidity - dew point- measurement methods –dry and wet bulb psychrometer- - sling psychrometer. - hair hygrometer – resistive hygrometer - dew cell - electrolytic hygrometers. moisture measurement – measurement of moisture in gases and liquids- capacitance hygrometer- impedance hygrometer. measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges. radiation measurement- types of radiations- geiger muller tube- ionization chamber- scintillation counter- applications of radiation detectors. apply different types of transducers for measurement of physical should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding moisture and radiation measurement Explain the construction and working principle of different types of humidity, 5 Understanding M2.02 moisture and radiation measuring techniques Compare and apply different transducers for Understanding M1.03 humidity, moisture and radiation 4 measurement Humidity measurement-Definition - relative humidity –Absolute humidity - Dew Point- Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers.

Aspect	What to prepare
	Moisture measurement – measurement of moisture in gases and liquids- capacitance hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges. Radiation measurement- Types of radiations- Geiger Muller tube- ionization chamber- scintillation counter- applications of radiation detectors. Apply different types of transducers for measurement of physical means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Understanding moisture and radiation measurement Explain the construction and working principle of different types of humidity, 5 Understanding M2.02 moisture and radiation measuring techniques Compare and apply different transducers for Understanding M1.03 humidity, moisture and radiation 4 measurement Humidity measurement-Definition - relative humidity –Absolute humidity - Dew Point- Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers. Moisture measurement – measurement of moisture in gases and liquids- capacitance hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges. Radiation measurement- Types of radiations- Geiger Muller tube- ionization chamber- scintillation counter- applications of radiation detectors. Apply different types of transducers for measurement of physical □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
humidity, moisture and radiation measuring instruments.
            Understand the fundamentals of humidity,
  M2.01                                                        5
Understanding
            moisture and radiation measurement
            Explain the construction and working
            principle of different types of humidity,          5
Understanding
  M2.02
            moisture and radiation measuring techniques
            Compare and apply different transducers for

Understanding
  M1.03     humidity, moisture and radiation                   4
            measurement
            Series Test – I                                    1

Contents:
Humidity measurement-Definition - relative humidity –Absolute humidity - Dew
Point-
Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair
hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers.
Moisture measurement – measurement of moisture in gases and liquids- capacitance
hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared
absorption or reflection moisture gauges- conductance or resistance moisture
gauges.
Radiation measurement- Types of radiations-
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

parameters like force, torque, speed and velocity measurements

This module is presented from the official syllabus scope for Industrial Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: parameters like force, torque, speed and velocity measurements
- Official duration: As specified
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.

5 Applying speed and velocity measurement. Explain the construction and working principle of different types of force, torque, 5 Understanding M3.02 speed and velocity measuring techniques. Compare and apply different transducers for force, torque, and speed and velocity 4 Applying M3.03 measurement. Force measurement- Hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer. Torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor. Speed and velocity measurement – revolution counter- magnetic drag or eddy current tacho meter- AC tachometer generator- DC tacho meter generator- - Stroboscope – translational velocity transducer. Compare and apply different types of transducers for acceleration,

5 Applying speed and velocity measurement. Explain the construction and working principle of different types of force, torque, 5 Understanding M3.02 speed and velocity measuring techniques. Compare and apply different transducers for force, torque, and speed and velocity 4 Applying M3.03 measurement. Force measurement- Hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer. Torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor. Speed and velocity measurement – revolution counter- magnetic drag or eddy current tacho meter- AC tachometer generator- DC tacho meter generator- - Stroboscope – translational velocity transducer. Compare and apply different types of transducers for acceleration, is an official topic in Module 3 of Industrial Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying speed and velocity measurement. Explain the construction and working principle of different types of force, torque, 5 Understanding M3.02 speed and velocity measuring techniques. Compare and apply different transducers for force, torque, and speed and velocity 4 Applying M3.03 measurement. Force measurement- Hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer. Torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor. Speed and velocity measurement – revolution counter- magnetic drag or eddy current

tacho meter- AC tachometer generator- DC tacho meter generator- - Stroboscope – translational velocity transducer. Compare and apply different types of transducers for acceleration, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying speed and velocity measurement. explain the construction and working principle of different types of force, torque, 5 understanding m3.02 speed and velocity measuring techniques. compare and apply different transducers for force, torque, and speed and velocity 4 applying m3.03 measurement. force measurement- hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer. torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor. speed and velocity measurement – revolution counter- magnetic drag or eddy current tacho meter- ac tachometer generator- dc tacho meter generator- - stroboscope – translational velocity transducer. compare and apply different types of transducers for acceleration, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Applying speed and velocity measurement. Explain the construction and working principle of different types of force, torque, 5 Understanding M3.02 speed and velocity measuring techniques. Compare and apply different transducers for force, torque, and speed and velocity 4 Applying M3.03 measurement. Force measurement- Hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer. Torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor. Speed and velocity measurement – revolution counter- magnetic drag or eddy current tacho meter- AC tachometer generator- DC tacho meter generator- - Stroboscope – translational velocity transducer. Compare and apply different types of transducers for acceleration, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 5 Applying speed and velocity measurement. Explain the construction and working principle of different types of force, torque, 5 Understanding M3.02 speed and velocity measuring techniques. Compare and apply different transducers for force, torque, and speed and velocity 4 Applying M3.03 measurement. Force measurement- Hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer. Torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor. Speed and velocity measurement – revolution counter- magnetic drag or eddy current tacho meter- AC tachometer generator- DC tacho meter generator- - Stroboscope – translational velocity transducer. Compare and apply different types of transducers for acceleration, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
parameters like force, torque, speed and velocity measurements
            Understand the fundamentals of force, torque,
  M3.01                                                        5
Applying
            speed and velocity measurement.
            Explain the construction and working
            principle of different types of force, torque,     5
Understanding
  M3.02
            speed and velocity measuring techniques.

            Compare and apply different transducers for
            force, torque, and speed and velocity              4
Applying
  M3.03
            measurement.

Contents:
Force measurement- Hydraulic load cells – pneumatic load cell – strain gauge load
cell-
electric force transducer.
Torque measurement- rotating torque sensor- stationary torque sensor- proximity
torque
sensor.
Speed and velocity measurement – revolution counter- magnetic drag or eddy
current tacho
meter- AC tachometer generator- DC tacho meter generator- - Stroboscope –
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

vibration, thickness, and miscellaneous measurements.

This module is presented from the official syllabus scope for Industrial Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: vibration, thickness, and miscellaneous measurements.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

vibration, thickness and miscellaneous 5 Understanding measurements. Explain the construction and working principle of different types of acceleration,

vibration, thickness and miscellaneous 5 Understanding measurements. Explain the construction and working principle of different types of acceleration, is an official topic in Module 4 of Industrial Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify vibration, thickness and miscellaneous 5 Understanding measurements. Explain the construction and working principle of different types of acceleration, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, vibration, thickness and miscellaneous 5 understanding measurements. explain the construction and working principle of different types of acceleration, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What vibration, thickness and miscellaneous 5 Understanding measurements. Explain the construction and working principle of different types of acceleration, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ vibration, thickness and miscellaneous 5 Understanding measurements. Explain the construction and working principle of different types of acceleration, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

vibration, thickness, and miscellaneous measuring techniques. Compare and apply different transducers for

vibration, thickness, and miscellaneous measuring techniques. Compare and apply different transducers for is an official topic in Module 4 of Industrial Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify vibration, thickness, and miscellaneous measuring techniques. Compare and apply different transducers for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, vibration, thickness, and miscellaneous measuring techniques. compare and apply different transducers for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What vibration, thickness, and miscellaneous measuring techniques. Compare and apply different transducers for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ vibration, thickness, and miscellaneous measuring techniques. Compare and apply different transducers for □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

acceleration, vibration, thickness, and 5 Applying miscellaneous measurements. Acceleration and vibration measurement- Seismic accelerometer- LVDT accelerometer - Piezoelectric accelerometer – eddy current proximity sensor as vibration pick up- mechanical vibration sensors. Thickness measurement- inductive methods- eddy current transducer method- capacitive method- ultrasonic method Miscellaneous measurements using Advanced sensors- Smoke& gas detector, leak detectors, flame detectors, Nanosensors, MEMS. Text / Reference: T/R Book Title /Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai and Sons. R2 S.K Singh, Industrial Instrumentation and Control, TMH. R3 R.K Jain, Mechanical and Industrial measurements, Khanna Publishers. R4 Krishnaswami, Industrial instrumentation, New Age Publications. Online resources: SL.No Website Link 1 www.reference.com › Science › Physics › Motion & Mechanics 2 https://study.com/.../what-is-acceleration-definition-and-formula.html

acceleration, vibration, thickness, and 5 Applying miscellaneous measurements. Acceleration and vibration measurement- Seismic accelerometer- LVDT accelerometer - Piezoelectric accelerometer – eddy current proximity sensor as vibration pick up- mechanical vibration sensors. Thickness measurement- inductive methods- eddy current transducer method- capacitive method- ultrasonic method Miscellaneous measurements using Advanced sensors- Smoke& gas detector, leak detectors, flame detectors, Nanosensors, MEMS. Text / Reference: T/R Book Title /Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai and Sons. R2 S.K Singh, Industrial Instrumentation and Control, TMH. R3 R.K Jain, Mechanical and Industrial measurements, Khanna Publishers. R4 Krishnaswami, Industrial instrumentation, New Age Publications. Online resources: SL.No Website Link 1 www.reference.com › Science › Physics › Motion & Mechanics 2 https://study.com/.../what-is-acceleration-definition-and-formula.html is an official topic in Module 4 of Industrial Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify acceleration, vibration, thickness, and 5 Applying miscellaneous measurements. Acceleration and vibration measurement- Seismic accelerometer- LVDT accelerometer - Piezoelectric accelerometer – eddy current proximity sensor as vibration pick up- mechanical vibration sensors. Thickness measurement- inductive methods- eddy current transducer method- capacitive method- ultrasonic method Miscellaneous measurements using Advanced sensors- Smoke& gas detector, leak detectors, flame detectors, Nanosensors, MEMS. Text / Reference: T/R Book Title /Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai and Sons. R2 S.K Singh, Industrial Instrumentation and Control, TMH. R3 R.K Jain, Mechanical and Industrial measurements, Khanna Publishers. R4 Krishnaswami, Industrial instrumentation, New Age Publications. Online resources: SL.No Website Link 1 www.reference.com › Science › Physics › Motion & Mechanics 2 https://study.com/.../what-is-acceleration-definition-and-formula.html using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, acceleration, vibration, thickness, and 5 applying miscellaneous measurements. acceleration and vibration measurement- seismic accelerometer- lvdt accelerometer - piezoelectric accelerometer – eddy current proximity sensor as vibration pick up- mechanical vibration sensors. thickness measurement- inductive methods- eddy current transducer method- capacitive method- ultrasonic method miscellaneous measurements using advanced sensors- smoke& gas detector, leak detectors, flame detectors, nanosensors, mems. text / reference: t/r book title /author a.k. sawhney, a course in mechanical measurements and instrumentation, r1 dhanpat rai and sons. r2 s.k singh, industrial instrumentation and control, tmh. r3 r.k jain, mechanical and industrial measurements, khanna publishers. r4 krishnaswami, industrial instrumentation, new age publications. online resources: sl.no website link 1 www.reference.com › science › physics › motion & mechanics 2 https://study.com/.../what-is-acceleration-definition-and-formula.html should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What acceleration, vibration, thickness, and 5 Applying miscellaneous measurements. Acceleration and vibration measurement- Seismic accelerometer- LVDT accelerometer - Piezoelectric accelerometer – eddy current proximity sensor as vibration pick up- mechanical vibration sensors. Thickness measurement- inductive methods- eddy current transducer method- capacitive method- ultrasonic method Miscellaneous measurements using Advanced sensors- Smoke& gas detector, leak detectors, flame detectors, Nanosensors, MEMS. Text / Reference: T/R

Aspect	What to prepare
	Book Title /Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai and Sons. R2 S.K Singh, Industrial Instrumentation and Control, TMH. R3 R.K Jain, Mechanical and Industrial measurements, Khanna Publishers. R4 Krishnaswami, Industrial instrumentation, New Age Publications. Online resources: SL.No Website Link 1 www.reference.com › Science › Physics › Motion & Mechanics 2 https://study.com/.../what-is-acceleration-definition-and-formula.html means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] acceleration, vibration, thickness, and 5 Applying miscellaneous measurements. Acceleration and vibration measurement- Seismic accelerometer- LVDT accelerometer - Piezoelectric accelerometer – eddy current proximity sensor as vibration pick up- mechanical vibration sensors. Thickness measurement- inductive methods- eddy current transducer method- capacitive method- ultrasonic method Miscellaneous measurements using Advanced sensors- Smoke& gas detector, leak detectors, flame detectors, Nanosensors, MEMS. Text / Reference: T/R Book Title /Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai and Sons. R2 S.K Singh, Industrial Instrumentation and Control, TMH. R3 R.K Jain, Mechanical and Industrial measurements, Khanna Publishers. R4 Krishnaswami, Industrial instrumentation, New Age Publications. Online resources: SL.No Website Link 1 www.reference.com › Science › Physics › Motion & Mechanics 2 https://study.com/.../what-is-acceleration-definition-and-formula.html [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
vibration, thickness, and miscellaneous measurements.
              Understand the fundamentals of acceleration,
   M4.01      vibration, thickness and miscellaneous                5
Understanding
              measurements.
              Explain the construction and working
              principle of different types of acceleration,
                                                                    5
Understanding
   M4.02      vibration, thickness, and miscellaneous
              measuring techniques.

              Compare and apply different transducers for
   M4.03      acceleration, vibration, thickness, and               5
Applying
              miscellaneous measurements.
              Series Test – 2                                       1

 Contents:
 Acceleration and vibration measurement- Seismic accelerometer- LVDT
accelerometer -
 Piezoelectric accelerometer – eddy current proximity sensor as vibration pick
up-
 mechanical vibration sensors.
 Thickness measurement- inductive methods- eddy current transducer method-
capacitive
 method- ultrasonic method
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain density, Specific gravity, and pH 5 Understanding measurement Explain the construction and working principle of different types of Viscosity,. (3 marks)

Answer: Begin with the standard definition or identity of *density, Specific gravity, and pH 5 Understanding measurement Explain the construction and working principle of different types of Viscosity,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding density, Specific gravity, and pH measuring instruments. Compare and apply different transducers for. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding density, Specific gravity, and pH measuring instruments. Compare and apply different transducers for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Viscosity, density, Specific gravity, and pH 5 Understanding measurements Viscosity measurement- Absolute viscosity - Kinematic viscosity - relative viscosity – units- capillary viscometer- Saybolt Viscometer - Redwood Viscometer - Applications of viscometer. Density and Specific gravity measurement- Definition- hydrometer- density measurement using LVDT- force balance method- differential pressure method. pH Measurements -Sorensen's scale–electrodes for pH measurements- Hydrogen electrode, glass electrode- calomel electrode, combined pH electrode- Digital pH meter. Explain the construction, working principle and applications of. (3 marks)

Answer: Begin with the standard definition or identity of *Viscosity, density, Specific gravity, and pH 5 Understanding measurements Viscosity measurement- Absolute viscosity - Kinematic viscosity - relative viscosity – units- capillary viscometer- Saybolt Viscometer - Redwood Viscometer - Applications of viscometer. Density and Specific gravity measurement- Definition- hydrometer- density measurement using LVDT- force balance method- differential pressure method. pH Measurements -Sorensen's scale–electrodes for pH measurements- Hydrogen electrode, glass electrode- calomel electrode, combined pH electrode- Digital pH meter. Explain the construction, working principle and applications of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

> □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 5 Understanding moisture and radiation measurement Explain the construction and working principle of different types of humidity, 5 Understanding M2.02 moisture and radiation measuring techniques Compare and apply different transducers for Understanding M1.03 humidity, moisture and radiation 4 measurement Humidity measurement-Definition - relative humidity –Absolute humidity - Dew Point- Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers. Moisture measurement – measurement of moisture in gases and liquids- capacitance hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges. Radiation measurement- Types of radiations- Geiger Muller tube- ionization chamber- scintillation counter- applications of radiation detectors. Apply different types of transducers for measurement of physical. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding moisture and radiation measurement Explain the construction and working principle of different types of humidity, 5 Understanding M2.02 moisture and radiation measuring techniques Compare and apply different transducers for Understanding M1.03 humidity, moisture and radiation 4 measurement Humidity measurement-Definition - relative humidity –Absolute humidity - Dew Point- Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers. Moisture measurement – measurement of moisture in gases and liquids- capacitance hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges. Radiation measurement- Types of radiations- Geiger Muller tube- ionization chamber- scintillation counter- applications of radiation detectors. Apply different types of transducers for measurement of physical*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 5 Applying speed and velocity measurement. Explain the construction and working principle of different types of force, torque, 5 Understanding M3.02 speed and velocity measuring techniques. Compare and apply different transducers for force, torque, and speed and velocity 4 Applying M3.03 measurement. Force measurement- Hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer. Torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor. Speed and velocity measurement – revolution counter- magnetic drag or eddy current tacho meter- AC tachometer generator- DC tacho meter generator- - Stroboscope – translational velocity transducer. Compare and apply different types of transducers for acceleration,. (3 marks)

Answer: Begin with the standard definition or identity of *5 Applying speed and velocity measurement. Explain the construction and working principle of different types of force, torque, 5 Understanding M3.02 speed and velocity measuring techniques. Compare and apply different transducers for force, torque, and speed and velocity 4 Applying M3.03 measurement. Force measurement- Hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer. Torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor. Speed and velocity measurement – revolution counter- magnetic drag or eddy current tacho meter- AC tachometer generator- DC tacho meter generator- - Stroboscope – translational velocity transducer. Compare and apply different types of transducers for acceleration,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain vibration, thickness and miscellaneous 5 Understanding measurements. Explain the construction and working principle of different types of acceleration,. (3 marks)

Answer: Begin with the standard definition or identity of *vibration, thickness and miscellaneous 5 Understanding measurements. Explain the construction and working principle of different types of acceleration,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain vibration, thickness, and miscellaneous measuring techniques. Compare and apply different transducers for. (3 marks)

Answer: Begin with the standard definition or identity of *vibration, thickness, and miscellaneous measuring techniques. Compare and apply different transducers for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain acceleration, vibration, thickness, and 5 Applying miscellaneous measurements. Acceleration and vibration measurement- Seismic accelerometer- LVDT accelerometer - Piezoelectric accelerometer – eddy current proximity sensor as vibration pick up- mechanical vibration sensors. Thickness measurement- inductive methods- eddy current transducer method- capacitive method- ultrasonic method Miscellaneous measurements using Advanced sensors- Smoke& gas detector, leak detectors, flame detectors, Nanosensors, MEMS. Text / Reference: T/R Book Title /Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai and Sons. R2 S.K Singh, Industrial Instrumentation and Control, TMH. R3 R.K Jain, Mechanical and Industrial measurements, Khanna Publishers. R4 Krishnaswami, Industrial instrumentation, New Age Publications. Online resources: SL.No Website Link 1 www.reference.com › Science › Physics › Motion & Mechanics 2 https://study.com/.../what-is-acceleration-definition-and-formula.html. (3 marks)

Answer: Begin with the standard definition or identity of *acceleration, vibration, thickness, and 5 Applying miscellaneous measurements. Acceleration and vibration measurement- Seismic accelerometer- LVDT accelerometer - Piezoelectric accelerometer – eddy current proximity sensor as vibration pick up- mechanical vibration sensors. Thickness measurement- inductive methods- eddy current transducer method- capacitive method- ultrasonic method Miscellaneous measurements using Advanced sensors- Smoke& gas detector, leak detectors, flame detectors, Nanosensors, MEMS. Text / Reference: T/R Book Title /Author A.K. Sawhney, A course in Mechanical* *measurements and Instrumentation, R1 Dhanpat Rai and Sons. R2 S.K Singh, Industrial Instrumentation and Control, TMH. R3 R.K Jain, Mechanical and Industrial measurements, Khanna Publishers. R4 Krishnaswami, Industrial instrumentation, New Age Publications. Online resources: SL.No Website Link 1 www.reference.com › Science › Physics › Motion & Mechanics 2 https://study.com/.../what-is-acceleration-definition-and-formula.html*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **density, Specific gravity, and pH 5 Understanding measurement Explain the construction and working principle of different types of Viscosity,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **5 Understanding moisture and radiation measurement Explain the construction and working principle of different types of humidity, 5 Understanding M2.02 moisture and radiation measuring techniques Compare and apply different transducers for Understanding M1.03**

humidity, moisture and radiation 4 measurement Humidity measurement-Definition - relative humidity –Absolute humidity - Dew Point- Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers. Moisture measurement – measurement of moisture in gases and liquids-capacitance hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges. Radiation measurement- Types of radiations-Geiger Muller tube- ionization chamber- scintillation counter- applications of radiation detectors. Apply different types of transducers for measurement of physical.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **5 Applying speed and velocity measurement. Explain the construction and working principle of different types of force, torque, 5 Understanding M3.02 speed and velocity measuring techniques. Compare and apply different transducers for force, torque, and speed and velocity 4 Applying M3.03 measurement. Force measurement-Hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer. Torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor. Speed and velocity measurement – revolution counter- magnetic drag or eddy current tacho meter- AC tachometer generator- DC tacho meter generator- - Stroboscope – translational velocity transducer. Compare and apply different types of transducers for acceleration,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **vibration, thickness and miscellaneous 5 Understanding measurements. Explain the construction and working principle of different types of acceleration,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- SL.No Website Link
- www.reference.com › Science › Physics › Motion & Mechanics
- https://study.com/.../what-is-acceleration-definition-and-formula.html

IN-PAGE SEARCH

Search this lesson